Letter of Intent and Previous Work

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My interests include mathematical modeling via
convex and combinatorial optimization, reinforcement learning, probabilistic models, graph theory and dynamic programming algorithms. Interested in using probabilistic methods for creating suitable estimators and root cause analysis.
Currently researching various MCP-enabled RAG, Agentic, and GraphRAG architectures for text generation and their applicability in certain contexts relevant to Nike. Also working on a framework for evaluation of these workflows using built-in MLflow judges, standard evaluation metrics, and custom-tailored scorers and custom metrics.
Examples of my recent work and research interests are:

a) RAG and Agentic AI workflows and relevant vector retrieval techniques as discussed in this repo, this repo and this repo; a framework for evaluation of these workflows using built-in MLflow judges, standard evaluation metrics, and custom-tailored scorers and custom metrics – for details see the sample code here.
b) Causal Inference realized via agentic workflows – for details see markdown documents and sample code here.
c) Actor-Critic design pattern and related algorithms applied to agentic workflows – for details see the sample code here.
d) Reinforcement learning Policy Gradient algorithms such as PPO, applying those to agentic workflows mentioned in a). For the purpose used the `gymnasium` environment and `stable-baselines3` library. Ongoing work captured in this document.
e) Semantic Simulation framework : supplemental material, GitHub repo, HuggingFace family of model cards
f) Algorithms for Root Cause Analysis using Bayesian inference and Probabilistic Temporal Logic. For the purpose exploring the usability of causal inference algorithms included in the packages `causal-learn`, `causalml` and `dowhy`. For details refer to this document and this document.
g) Classical and Deep Learning algorithms for various image processing tasks such as synthetic noise generation, generative fill and shadow processing. Repos with code samples related to this work – smooth_gradient_outpaint and image_crop.
h) Redesign of the online Fulfillment algorithm using the ε-constraint method, replacing the weighted objectives method for scalarization , reformulating the problem as Mixed Integer optimization problem. For details refer to this document and this document.

My coding experience involve python, C++, C, Java.
Samples of my python code can be found here:

algorithms for smooth gradient outpainting
https://github.com/dimitarpg13/smooth_gradient_outpaint

image crop algorithm using object segmentation with the Dichotomous Segmentation Deep Learning model:
https://github.com/dimitarpg13/image_crop

Here are samples of my C++ code from past endeavors:
https://github.com/google/or-tools/compare/stable...dimitarpg13:ortools:dpg/PWL_solver_stable_py2.7_gtest_scipV6
https://github.com/dimitarpg13/testcode/blob/master/fraction.cpp
https://github.com/dimitarpg13/testcode/blob/master/fraction_mt.cpp
https://github.com/dimitarpg13/testcode/blob/master/fraction_bigint.cpp
https://github.com/dimitarpg13/cpp_testcode/tree/master/SudokuQlik/src

And here are relevant documents to software design, architecture, coding techniques and design

patterns:
https://github.com/dimitarpg13/BigIndex/blob/main/PresentationDGueorguiev2018.pdf
https://github.com/dimitarpg13/InsideTensorflow2Source/blob/master/Understanding_Tensorflow_2_source_code.pdf
https://github.com/dimitarpg13/UnderstandingPythonEcosystem
https://github.com/dimitarpg13/inside_cpp_object_model

And here are few repos about C++ language details and features:
https://github.com/dimitarpg13/cpp_effective_modern
https://github.com/dimitarpg13/cpp_move_semantics
https://github.com/dimitarpg13/cpp_templates_complete_guide
https://github.com/dimitarpg13/cpp_random_pieces